

Lecture 07: Advantage Actor Critic

Radoslav Neychev

References

These slides are almost the exact copy of Practical RL course week 6 slides.
Special thanks to YSDA team for making them publicly available.

Original slides link: [week06_policy_based](#)

REINFORCE algorithm

- Initialize NN weights $\theta_0 \leftarrow \text{random}$
- Loop:
 - Sample N sessions $\mathbf{z}$ under current $\pi_\theta(a|s)$
 - Evaluate policy gradient

$$\nabla J \approx \frac{1}{N} \sum_{i=0}^N \sum_{s, a \in \mathbf{z}_i} \nabla \log \pi_\theta(a|s) \cdot Q(s, a)$$

- Ascend $\theta_{i+1} \leftarrow \theta_i + \alpha \cdot \nabla J$

REINFORCE algorithm

- Initialize NN weights $\theta_0 \leftarrow \text{random}$

Q: is it off- or on-policy?

- Loop:
 - Sample N sessions $\mathbf{z}$ under current $\pi_\theta(a|s)$
 - Evaluate policy gradient

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REINFORCE algorithm

- Initialize NN weights $\theta_0 \leftarrow \text{random}$
- Loop:
 - Sample N sessions $\mathbf{z}$ under current $\pi_\theta(a|s)$
actions under current policy
= on-policy
 - Evaluate policy gradient

$$\nabla J \approx \frac{1}{N} \sum_{i=0}^N \sum_{s, a \in \mathbf{z}_i} \nabla \log \pi_\theta(a|s) \cdot Q(s, a)$$

- Ascend $\theta_{i+1} \leftarrow \theta_i + \alpha \cdot \nabla J$

value-based Vs policy-based

Value-based

- Q-learning, SARSA, MCTS value-iteration
- Solves harder problem
- Artificial exploration
- Learns from partial experience (temporal difference)
- Evaluates strategy for free :)

Policy-based

- REINFORCE, CEM
- Solves easier problem
- Innate exploration
- Innate stochasticity
- Support continuous action space
- Learns from full session only?



value-based Vs policy-based

Value-based

- Q-learning, SARSA, MCTS value-iteration
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Policy-based

- REINFORCE, CEM

We'll learn much more soon!

- Solves easier problem
- Innate exploration
- Innate stochasticity
- Support continuous action space
- ~~Learns from full session only~~



REINFORCE baselines

- Initialize NN weights $\theta_0 \leftarrow \text{random}$
- Loop:
 - Sample N sessions $\mathbf{z}$ under current $\pi_\theta(a|s)$
 - Evaluate policy gradient

$$\nabla J \approx \frac{1}{N} \sum_{i=0}^N \sum_{s, a \in \mathbf{z}_i} \nabla \log \pi_\theta(a|s) \cdot Q(s, a)$$

What is better for learning:
random action in good state
or
great action in bad state?

REINFORCE baselines

We can subtract arbitrary baseline $b(s)$

$$\nabla J = \mathop{E}_{\substack{s \sim p(s) \\ a \sim \pi_\theta(a|s)}} \nabla \log \pi_\theta(a|s) (Q(s, a) - b(s)) = \dots$$

$$\dots = \mathop{E}_{\substack{s \sim p(s) \\ a \sim \pi_\theta(a|s)}} \nabla \log \pi_\theta(a|s) Q(s, a) - \mathop{E}_{\substack{s \sim p(s) \\ a \sim \pi_\theta(a|s)}} \nabla \log \pi_\theta(a|s) b(s) = \dots$$

REINFORCE baselines

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Q: Can you simplify the second term?

Note that $b(s)$ does not depend on a

REINFORCE baselines

We can subtract arbitrary baseline $b(s)$

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$$\dots = E_{\substack{s \sim p(s) \\ a \sim \pi_{\theta}(a|s)}} \nabla \log \pi_{\theta}(a|s) Q(s, a) - E_{\substack{s \sim p(s) \\ a \sim \pi_{\theta}(a|s)}} \nabla \log \pi_{\theta}(a|s) b(s) = \dots$$

$$E_{\substack{s \sim p(s) \\ a \sim \pi_{\theta}(a|s)}} \nabla \log \pi_{\theta}(a|s) b(s) = b(s) \cdot E_{\substack{s \sim p(s) \\ a \sim \pi_{\theta}(a|s)}} \nabla \log \pi_{\theta}(a|s) = 0$$

REINFORCE baselines

We can subtract arbitrary baseline $b(s)$

$$\nabla J = E_{\substack{s \sim p(s) \\ a \sim \pi_{\theta}(a|s)}} \nabla \log \pi_{\theta}(a|s) (Q(s, a) - b(s)) = \dots$$

$$\dots = E_{\substack{s \sim p(s) \\ a \sim \pi_{\theta}(a|s)}} \nabla \log \pi_{\theta}(a|s) Q(s, a) - E_{\substack{s \sim p(s) \\ a \sim \pi_{\theta}(a|s)}} \nabla \log \pi_{\theta}(a|s) b(s) = \dots$$

Gradient direction doesn't change!

$$\dots = E_{\substack{s \sim p(s) \\ a \sim \pi_{\theta}(a|s)}} \nabla \log \pi_{\theta}(a|s) Q(s, a)$$

REINFORCE baselines

- Gradient direction ∇J stays the same
- Variance may change

Gradient variance: $Var[Q(s, a) - b(s)]$
as a random variable over (s, a)

$$Var[Q(s, a)] - 2 \cdot Cov[Q(s, a), b(s)] + Var[b(s)]$$

REINFORCE baselines

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 If $b(s)$ correlates with $Q(s, a)$, variance decreases

REINFORCE baselines

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as a random variable over (s, a)

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 Q: can you suggest any such $b(s)$?

REINFORCE baselines

- Gradient direction ∇J stays the same
- Variance may change

Gradient variance: $Var[Q(s, a) - b(s)]$
as a random variable over (s, a)

$$Var[Q(s, a)] - 2 \cdot Cov[Q(s, a), b(s)] + Var[b(s)]$$

Naive baseline: b = moving average Q
over all (s, a) , $Var[b(s)] = 0$, $Cov[Q, b] > 0$

REINFORCE baselines

Better baseline: $b(s) = V(s)$

$$\nabla J \approx \frac{1}{N} \sum_{i=0}^N \sum_{s, a \in z_i} \nabla \log \pi_{\theta}(a|s) \cdot (Q(s, a) - V(s))$$

Q: but how do we predict $V(s)$?

Actor-critic

- Learn both $V(s)$ and $\pi_{\theta}(a|s)$
- Hope for the best of both worlds :)



Advantage actor-critic

Idea: learn both $\pi_{\theta}(a|s)$ and $V_{\theta}(s)$

Use $V_{\theta}(s)$ to learn $\pi_{\theta}(a|s)$ faster!

Q: how can we estimate **$A(s,a)$**
from (s,a,r,s') and V-function?

Advantage actor-critic

Idea: learn both $\pi_{\theta}(a|s)$ and $V_{\theta}(s)$

Use $V_{\theta}(s)$ to learn $\pi_{\theta}(a|s)$ faster!

$$A(s, a) = Q(s, a) - V(s)$$

$$Q(s, a) = r + \gamma \cdot V(s')$$

$$A(s, a) = r + \gamma \cdot V(s') - V(s)$$

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Also: n-step
version

Advantage actor-critic

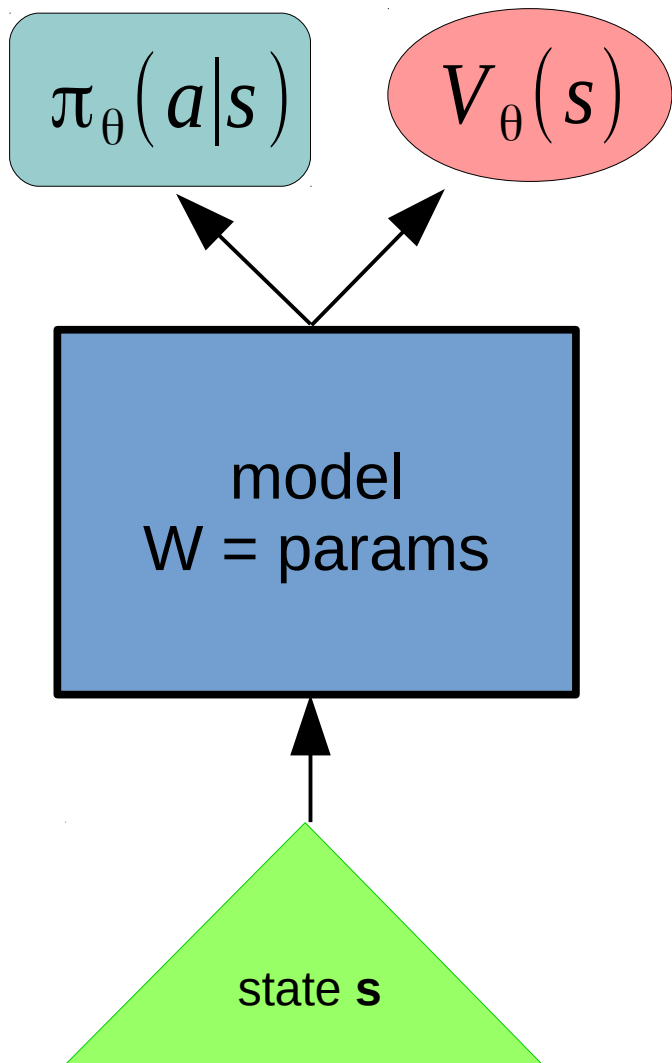
Idea: learn both $\pi_{\theta}(a|s)$ and $V_{\theta}(s)$

Use $V_{\theta}(s)$ to learn $\pi_{\theta}(a|s)$ faster!

$$A(s, a) = r + \gamma \cdot V(s') - V(s)$$

$$\nabla J_{actor} \approx \frac{1}{N} \sum_{i=0}^N \sum_{s, a \in z_i} \nabla \log \pi_{\theta}(a|s) \cdot \underbrace{A(s, a)}_{\text{consider const}}$$

Advantage actor-critic



Improve policy:

$$\nabla J_{actor} \approx \frac{1}{N} \sum_{i=0}^N \sum_{s,a \in z_i} \nabla \log \pi_{\theta}(a|s) \cdot A(s, a)$$

Improve value:

$$L_{critic} \approx \frac{1}{N} \sum_{i=0}^N \sum_{s,a \in z_i} (V_{\theta}(s) - [r + \gamma \cdot V(s')])^2$$

Continuous action spaces

What if there's continuously many actions?

- Robot control: control motor voltage
- Trading: assign money to equity

How does the algorithm change?

Continuous action spaces

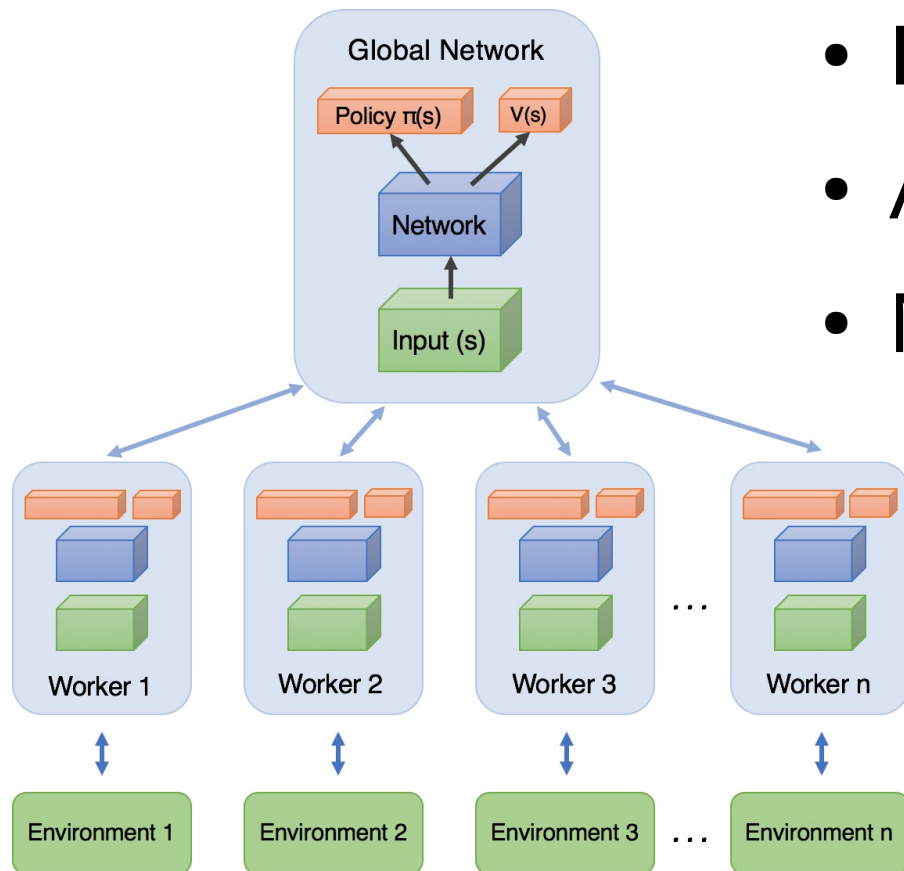
What if there's continuously many actions?

- Robot control: control motor voltage
- Trading: assign money to equity

How does the algorithm change?

it doesn't :)
Just plug in a different formula for
 $\pi(a|s)$, e.g. normal distribution

Asynchronous advantage actor-critic

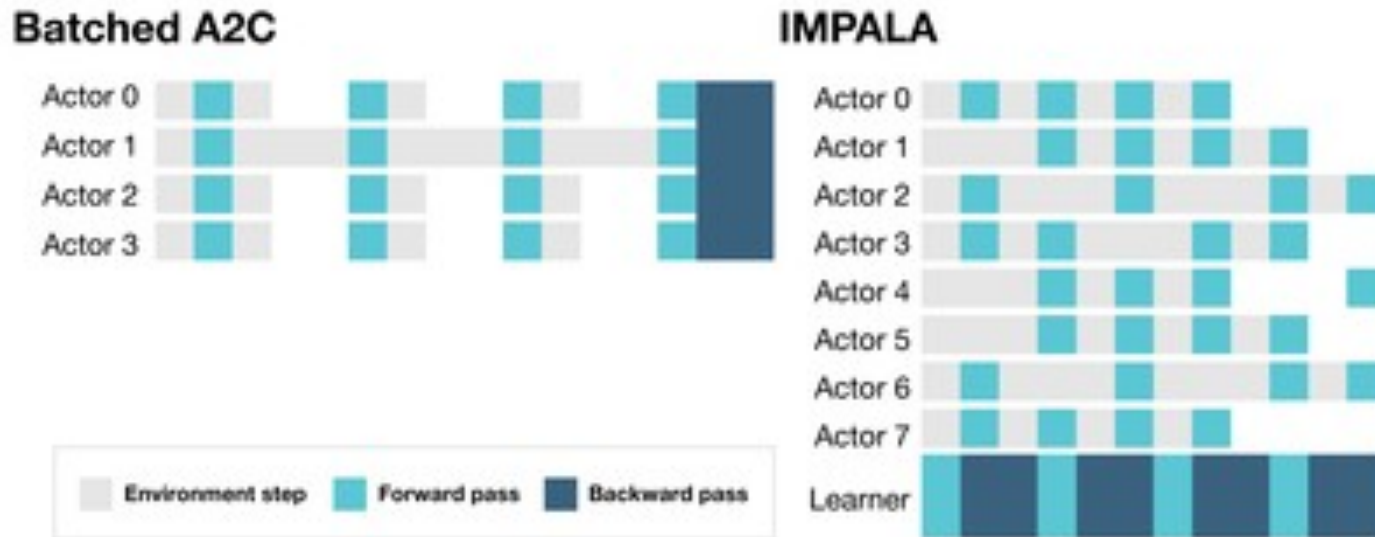


- Parallel game sessions
 - Async multi-CPU training
 - No experience replay
-
- LSTM policy
 - N-step advantage
 - No experience replay

Read more: <https://arxiv.org/abs/1602.01783>

IMPALA

- Massively parallel
- Separate actor / learner processes
 - Small experience replay w/ importance sampling



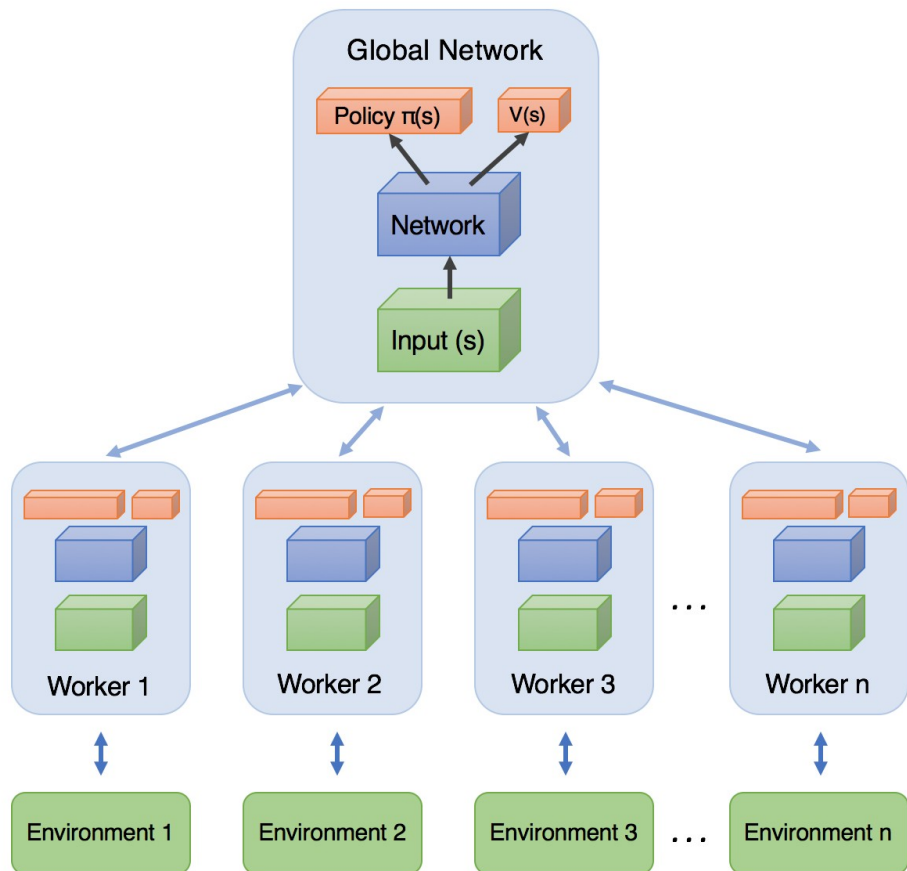
Read more: <https://arxiv.org/abs/1802.01561>

Duct tape zone

- $V(s)$ errors less important than in Q-learning
 - actor still learns even if critic is random, just slower
- Regularize with entropy
 - to prevent premature convergence
- Learn on parallel sessions
 - Or super-small experience replay
- Use logsoftmax for numerical stability



Asynchronous advantage actor-critic



- Most recent methods are using best from both Policy-based and Value-based worlds
- Remember the **Expected Grad-Log-Prob Lemma**
- See [this](#) paper (ICLR 2019) for the proof of the Variance reduction